SUSHREE SWATEEPRAJNYA BEHERA

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PROFESSIONAL SUMMARY

I am an experimental quantum hardware researcher with hands-on experience in designing, integrating, and characterizing complex photonic and atomic systems. I am proficient in laboratory automation, instrumentation control, subsystem interfacing, and building precision measurement platforms from scratch for high-stability environments. Alongside, I am experienced in navigating cleanroom fabrication workflows, managing multi-instrument data acquisition pipelines, and translating high-level physics requirements into scalable quantum hardware solutions.

Research Interest: Solid-state quantum photonics, quantum light sources, nanophotonic device integration, quantum networks, and precision optical measurement platforms.

RESEARCH POSITIONS

Quantum Sensing Researcher March 2026 - July 2026
QuBeats (M/s Nostradamus Technologies Private Limited), Hyderabad, India
In collaboration with Tata Institute of Fundamental Research Hyderabad

Researcher November 2024 - December 2025
Quantum Materials Team
14-month fixed term researcher for AQuaTSiC project
Luxembourg Institute of Science and Technology, Esch-sur-Alzette
In association with Université du Luxembourg, Luxembourg

Guest Scientist May 2024 - July 2024
Quantum Photonics Group - Experimental Physics
Universität des Saarlandes, Saarbrücken, Germany

Project Associate I October 2022 - April 2024
Indian Institute of Science, Bengaluru, India
Quantum Optics and Quantum Information Processing Group
Department of Instrumentation and Applied Physics
Center of Excellence in Quantum Technology

Visiting researcher July 2022 - October 2022
Department of Physics
Institute of Mathematical Sciences, Chennai, India

EDUCATION

Indian Institute of Science Education and Research Tirupati, India
BS-MS Dual Degree

August 2016 – August 2021
Postgraduation CGPA (MS): 8.3/10

MS Thesis Work: "Quantum Volume: Capability of a quantum computer and effective implementation of quantum circuits"

- Computed noise in state preparation, gate operations, and measurements that limits a quantum processor performance.
- Analysed a protocol for accurate estimation of quantum volume in noisy NISQ devices.

RESEARCH PROJECTS

QuBeats (M/s Nostradamus Technologies Private Limited), India March 2026 – July 2026
Quantum Sensing Researcher

Warm atomic vapour experiments - optically pumped magnetometers with Rubidium (*Rb*) atoms.

- Designed and built a python-based Graphical User Interface (GUI) for compact optically pumped magnetometer to sense and measure the magnetic field of earth.
- Performed data acquisition and analysis of frequency-modulated signals in an ^{85}Rb-enriched (OTS-coated) cell, nonlinear magneto-optical rotation (NMOR) experiment housed within a Twinleaf mu-metal magnetic shield; integrated the Moku:Go platform for waveform generation, lock-in detection (SRS SR830 DSP), and oscilloscope measurements.
- Built a python-based digital lock-in amplifier for signal extraction and phase-sensitive detection using butterworth low-pass filtering.
- Designed, wound, and characterized 3-axis Helmholtz coils for precision magnetic-field generation and control in quantum optics experiments.

Luxembourg Institute of Science and Technology, Luxembourg November 2024 – December 2025
Researcher
In association with Université du Luxembourg, Luxembourg

Two-photon interference using color centers (divacancy) in silicon carbide (4H-SiC).

- Designed and built an optical fiber gluing station for waveguide–fiber interfacing from scratch, enabling stable chip–fiber coupling.
- Integrated and automated 6-axis stages and visible to near-infrared (780–1220 nm) laser systems (Thorlabs SLDs & Cobolt / HÜBNER Photonics & Roithner Lasertechnik) via python with 4H-SiC nanophotonic devices.
- Led cleanroom fabrication of specialty fibers (1060XP and ZBLAN in association with Flawless Photonics Inc.), encompassing documentation, process risk assessment (PRA), and logistics, including process optimization, execution, quality control, characterization using field-emission scanning electron microscopy (FE-SEM; Hitachi SU-70) and ultralow-loss fiber splicing (0.01 dB).
- Evaluated, qualified, and procured optical adhesives and UV-curing system (Dymax BlueWave® 200 version 3.1) for photonic chip-level fiber attachment, defining bonding protocols and equipment requirements for long-term mechanical and optical stability.
- Conducted theoretical study of neutral divacancy (VV) in 4H-SiC, analyzing electronic structure, and optical transitions involved in single-photon generation.

Universität des Saarlandes, Germany
May 2024 – July 2024
Guest Scientist

Lasing from Cold Ytterbium Atoms, Quantum Communication with Calcium (Ca^+) ions and Quantum Photonics.

- Investigated coherent population trapping and dark-state formation in a Λ-system of ^{174}Yb using 680 nm and 770 nm repump lasers; performed fluorescence spectroscopy on the $^1S_0$ to $^3S_1$ transition via $^3P_1$.
- Built a double pass AOM (Acousto-Optic Modulator; Brimrose TEF-125-100-397) with a 125 MHz central frequency setup for 397nm laser and data acquisition, analysis, and stability characterization of pulsed laser with varying radio-frequency using object oriented programming with MATLAB.

Indian Institute of Science, India
October 2022 - April 2024
Project Associate I

Quantum hyperentanglement (polarization and path degrees of freedom) with spontaneous parametric down-conversion (SPDC) using type I and type II β-BaB_2O_4 (BBO) crystals and quantum entanglement using Type II SPDC with PPKTP Crystal.

- Generation of entangled photon pairs and heralded single photon sources, detection and estimation by EMCCD camera to establish the violation of Bell's inequalities.
- Numerical simulation suite to map 2D phase-space topographies of the CHSH inequality, visually demonstrating Tsirelson's bound ($2\sqrt{2}$) and rotational invariance in quantum entanglement.

Institute of Mathematical Sciences (IMSc), India
July 2022 – October 2022
Visiting Researcher

- Topological networks for quantum communication between distant qubits, robust networks for efficient communication.

LABORATORY HARDWARE EXPERIENCE

Lasers & Optics Devices: CW lasers (visible to near-infrared; ONDAX SureLock, Coherent, Thorlabs SLDs, Cobolt / HÜBNER Photonics & Roithner Lasertechnik), non-linear optics (Newlight; NCBBO5300-405(II)-HA5 and NCBBO5100-405(I)-HA3), free-space and fiber optics, optical components (Thorlabs), RF drivers, fluorescence spectroscopy, Acousto-Optic Modulator (AOM; Brimrose TEF-125-100-397), fiber splicing (Fujikura 90S), spectrometer (SpectraPro HRS 750 equipped with a NIRvana HS camera, Teledyne Princeton Instruments and Ocean Insights QE Pro), UV-curing system (Dymax BlueWave® 200 version 3.1)

Detectors & Timing: Single-photon detectors (SPCM-AQ4C, APDs), Single-Photon Avalanche Diode (SPAD; Excelitas SPCM-800-44-FC), Photomultiplier tube (PMT; Hamamatsu H9656), Electron-Multiplied Charge-Coupled Device (EMCCD camera; Oxford Instruments, Andor iXon), time-correlated single-photon counting (TCSPC), Time Tagger (Swabian Instruments), PicoQuant HydraHarp series (Hydra II), PicoHarp (TCSPC), and High-gain balanced photodetector (PD10B)

Automation & Electronic Instrumentations: Hardware automation and synchronization of lasers with python and MATLAB, CMOS camera (Thorlabs; CS126CU), 6-axis NanoMax stages (Thorlabs - 600/M series), Digital oscilloscopes (PicoScope 5444D PC (4-Channel, 200 MHz, 1 GSPS, Mixed Signal Oscilloscope), Digilent AD3), Moku:Go (Liquid Instruments)), lock-in amplifiers (Stanford Research Systems - SR830 DSP), Red Pitaya (STEMlab 125-14), waveform generators, RF drivers, current sources and precision magnetic-field controllers (Twinleaf systems), Low-Pass Electrical Filter $\leq$ 100 kHz (Thorlabs - EF502)

Cleanroom Experience (ISO 4 and ISO 5) & Characterization: Specialty fiber fabrication (1060XP, ZBLAN), nanophotonic device processing workflows and fiber–chip integration techniques, nanostructure characterization using FE-SEM (Hitachi SU-70), and laboratory safety and process protocol compliance in controlled environments.

ADDITIONAL SKILLS

Programming Language	Python (NumPy, SciPy, pandas), HTML, LaTeX
Quantum SDK	Qiskit
Software	Mathematica, MATLAB, Overleaf, OriginPro
Platforms	AWS Braket (SV1 simulator, hardware backends), IBM Quantum Experience Jupyter, PyCharm, VS Code, Git

PUBLICATIONS

Observation of geometric phase in a molecular Aharonov-Bohm system using IBM Quantum Computer
Gaurav Rudra Malik, *Sushree Swateeprajnya Behera*, Shubham Kumar, Bikash K. Behera, Prasanta K. Panigrahi
Preprint: arXiv:1909.00298 [quant-ph], 2019
https://doi.org/10.48550/arXiv.1909.00298

Controlling asymptotic self-similar optical beams in a semiconductor waveguide doped with quantum dots
Sushree Swateeprajnya Behera, Nikhitha Nunavath, Vineetha Ashok, Thokala Soloman Raju
Published as conference paper at Optical Society of America (OSA), PHOTONICS 2018
© 2018 Optical Society of America *OCIS codes: 190.4370, 060.5530, 060.2320.*
https://doi.org/10.13140/RG.2.2.27027.03366

INVITED TALKS AND PRESENTATIONS

Two-photon interference with divacancies in silicon carbide
Invited talk - oral presentation
Quantum SiC Workshop, Stralsund, Germany (October 2025)

Towards two-photon interference with color centers in silicon carbide
Oral and poster presentation
Greenhorn Meeting, Saarbrücken, Germany (September 2025)

Multiplexing divacancies in silicon carbide for quantum networks
Poster presentation
DPPM PhD Day, University of Luxembourg, Luxembourg (April 2025)

Multiplexing color centers in silicon carbide for quantum networks
Poster presentation
DPG Conference, Bonn, Germany (March 2025)

Journal club presentation in quantum metrology
Oral presentation - Studied the optimal bound for quantum precision enhancement compared to the classical scenario
IISC Bengaluru, India (June 2023)

SUMMER SCHOOLS

Qubit by Qubit, Introduction to Quantum Computing Course (October 2020 - May 2021)
IBM quantum & QISKIT Community, The Coding School

Quantum Information and Quantum Technology (QIQT-2019)
Summer School, IISER Kolkata

ACADEMIC ACHIEVEMENTS

Graduate Aptitude Test in Engineering Physics GATE 2023
Qualified
Dept. of Science and Technology, Govt. of India

Winner at Poster presentation, International Conference on Fibre Optics and Photonics
(PHOTONICS 2018) IIT Delhi, India

ADDITIONAL EXPERIENCE

Open Day 2024: IISc Bengaluru
Volunteer Research Assistant: Laboratory Tour
Indian Institute of Science (IISC) Bengaluru, India

PHOTONICS 2023: International conference on fibre optics and photonics
Indian Institute of Science (IISC) Bengaluru, India
Conference Volunteer: Laboratory Tour

Open Day 2023: IISc Bengaluru
Volunteer Research Assistant: Laboratory Tour
Indian Institute of Science (IISC) Bengaluru, India

Students Conference on Photonics (SCOP) - 2022
PRL (Physical Research Laboratory), Ahmedabad, India

Certifications:

Fundamentals of Quantum Algorithms · Basics of Quantum Information · AWS Knowledge: Amazon Braket · Integrating Quantum and HPC · Quantum Diagonalization Algorithms · Variational Algorithm Design · Quantum Machine Learning · Foundations of Quantum Error Correction · General Formulation of Quantum Information · Quantum-Safe Cryptography · Quantum Business Foundations

PERSONAL DETAILS

Name: Sushree Swateeprajnya Behera

Date of Birth: April 05th 1998

Citizenship / Birth Place: Indian / Odisha, India

Gender: Female (She)

Extracurricular Activities: Swimming, Ice Skating, Hiking, Painting, Culinary Arts and Cooking

Alternate Email IDs: sushree.s.behera@outlook.com & sushswatee@gmail.com